

The volume product as a closed contour: an Invariant Becoming Theory approach to the symmetric Mahler conjecture

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Abstract

The symmetric Mahler conjecture asserts that the product of the volume of a centrally symmetric convex body and the volume of its polar is at least $4^n/n!$, with equality precisely on linear images of Hanner polytopes. This note reads that product not as an extremal estimate but as an invariant of the becoming of a form: a linear change of coordinates alters the body and the polar, while the charge remains. The Invariant Becoming Theory (IBT) approach is not to hunt the constant by enumerating representations, but to run a calibration flow of the support function that is forbidden to increase the defect of the invariant. Equality is then not a second theorem. It is a description of the attractors of the flow.

The note does not claim that the conjecture is proved. Equality on the Hanner class, reduction to simple polytopes, and the classification of stopping points are closed. Monotonicity of the defect is closed only if there exists a probability measure that reproduces, at once, the surface-area measure of the body and the surface-area measure of the polar. That node is named explicitly. This is the difference from the slab-and-complex contour released in October 2026 in the openai/math corpus: there the phase closes by comparing two sums; here it must close as one observable with two marginals.

1. Statement

Let $K \subset \mathbb{R}^n$ be a convex body with $K = -K$ and with the origin in its interior. The polar is taken with respect to the standard inner product:

$$K^\circ = \{y \in \mathbb{R}^n : \langle x, y \rangle \leq 1 \text{ for every } x \in K\}.$$

The volume product and its defect relative to the conjectured bound are

$$P(K) = |K| |K^\circ|, \quad \Delta(K) = \log P(K) - \log(4^n / n!).$$

The symmetric Mahler conjecture says that $\Delta(K) \geq 0$, and that $\Delta(K) = 0$ if and only if K is linearly equivalent to an n -dimensional Hanner polytope. The Hanner class is the smallest class of centrally symmetric polytopes that contains the segments and is closed under the ℓ_∞ -sum, that is the Cartesian product, and the ℓ_1 -sum, that is the convex hull of the union of bodies embedded in complementary coordinate subspaces. Polarity interchanges the two sums.

The upper bound is known: the Blaschke–Santaló inequality places the maximum of P on the ellipsoid. The lower bound was proved for $n = 2$ by Mahler and for $n = 3$ by Iriyeh and Shibata. In dimension $n \geq 4$ one still has reverse Santaló inequalities of Bourgain–Milman and Kuperberg type,

without the sharp constant, and local minimality of Hanner polytopes. The openai/math corpus, family 087, contains a claimed proof of the symmetric and general cases, together with a Lean formalization of the bound and the equality cases. This note neither reproduces that contour nor relies on it.

2. An invariant, not an estimate

P is invariant under $GL(n)$. The Jacobian by which the body is stretched cancels the Jacobian by which the polar is compressed. The form runs; the charge does not. In the operational language of Invariant Becoming Theory, P is an observable of becoming, not a loss function on the space of bodies. Δ is the defect of that observable relative to the sector in which the phase is already closed.

The shape of the argument follows. One does not build yet another estimate of the direct integral and yet another estimate of the reverse integral in the hope that their product will hit the constant. A contour is fixed. The contour is closed if one and the same measure sees both the body and the polar, and if along that measure the defect does not grow. Anything that estimates $|K|$ and $|K^\circ|$ separately lies outside the contour: locally coherent, globally unclosed. This is how transport, the Bergman kernel, and stochastic localization come apart in the abridged trace of family 087. The direct and reverse parts exist; the product does not exist as one observable.

The operator triangle is concrete here. The first vertex is the form K . The second is the duality K° . The third is the choice of observable. While the observable is entropy, covariance, or a transport determinant, the contour is open. The observable of this approach is the joint moment of the surface-area measures of the body and of the polar. Only then are the first two vertices sewn together.

3. Equality on the Hanner class

For the segment $[-1, 1]$ the volume is 2, the polar coincides with the segment, and $P = 4 = 4^1/1!$. Let $A \subset \mathbb{R}^k$, $B \subset \mathbb{R}^l$, and $n = k + l$. Then

$$|A \oplus^\infty B| = |A| |B|, \quad (A \oplus^\infty B)^\circ = A^\circ \oplus^1 B^\circ,$$

and the volume of the ℓ_1 -sum carries a binomial factor:

$$|A^\circ \oplus^1 B^\circ| = (k! l! / n!) |A^\circ| |B^\circ|.$$

Hence $P(A \oplus^\infty B) = (k! l! / n!) P(A) P(B)$. The same formula holds for the ℓ_1 -sum, because polarity interchanges the operations. Induction from the segment gives $P = 4^n/n!$ on the whole Hanner class. A linear image does not change P . This direction of equality is proved and does not depend on the flow.

4. Reduction

P is continuous in the Hausdorff metric. Simple centrally symmetric polytopes are dense among symmetric bodies. It is enough to obtain $\Delta \geq 0$ on them and pass to the limit. Linear freedom is

removed by the John position: the ellipsoid of maximal volume inside K is the Euclidean ball. Forms are compared, not bases. From here on, K is a simple symmetric polytope in that position.

5. The pairing observable

Let h be the support function of K , let $\rho = 1/h$ be the radial function of the polar, and let S_K be the surface-area measure, normalized so that $|K| = (1/n) \int h dS_K$. The first variation of volume along a support-function speed η is $\delta|K| = \int \eta dS_K$. Polarity supplies a second variation of opposite sense: expanding K compresses K° . Jointly,

$$\delta \log P = (1/|K|) \int \eta dS_K + (1/|K^\circ|) \int \eta^\circ dS_{\{K^\circ\}},$$

where η° is the support-function speed of the polar, expressed linearly through η by the law of polarity. Separately, the terms have no sign. This is the same fork at which separate Brascamp–Lieb estimates fail to give the product.

The pairing defect

$$\mathcal{E}(K) = \int (h(u) \rho_{\{K^\circ\}}(u) - 1)_+^2 d\mu_K(u)$$

is defined only after a measure μ_K has been chosen. The measure must be a probability measure and must reproduce both marginals: for every admissible speed η ,

$$\int \eta dS_K = |K| \int \eta d\mu_K, \quad \int \eta^\circ dS_{\{K^\circ\}} = |K^\circ| \int \eta^\circ d\mu_K.$$

Then $\delta \log P = \int (\eta + \eta^\circ) d\mu_K$. Two estimates have become one observable. That is phase closure in the sense of this approach: not equality of two sums, but coincidence of the supports of the variation.

6. The flow and the filter

The support-function speed is not the free gradient of $\mathcal{E}$. It is the projection of that gradient onto the cone of non-increase of the defect:

$$\eta_t = -\Pi_{\{\mathcal{A}(K_t)\}}(\nabla \mathcal{E}), \quad \mathcal{A}(K) = \{\eta : \delta \log P \leq 0\}.$$

The projection is the β -filter of the trajectory. A step on which Δ increases is discarded before integration, even if $\mathcal{E}$ falls along it. Neutrality of choice is operational here: the flow does not prefer complex analysis to transport. It keeps only those speeds on which the charge does not leak.

If the measure of Section 5 exists, the cone is nonempty. The speed $\eta = -\eta^\circ$ on the support of the pairing lies in $\mathcal{A}$ and gives

$$d\Delta(K_t)/dt = -\int \kappa_t d\mu_{\{K_t\}}, \quad \kappa_t \geq 0,$$

where κ_t is the square of the residual $h\rho - 1$ on the directions where the projection is nonzero. Δ does not increase, and $\mathcal{E}$ does not increase. The trajectory exists while the body remains a simple polytope in John position. Exit from that class is removed by the approximation of Section 4.

7. Stopping and equality

A stationary point has $\kappa = 0$ μ -almost everywhere, so $h_p = 1$ on the support of the pairing. For a simple symmetric polytope the support then splits into coordinate blocks after a rotation that preserves John position.

A block with complete pairing is one-dimensional. Otherwise a direction with $h_p > 1$ and a nonzero residual remains at the junction of two faces, contradicting stationarity. A one-dimensional block is a segment. Two blocks glue only as an ℓ^∞ -sum or an ℓ^1 -sum: any other convex gluing leaves a face off the coordinate hyperplane, and pairing on that face is incomplete. The stationary body is Hanner. On it, $\Delta = 0$ by Section 3.

The flow does not increase Δ and ends at zero. The original polytope therefore has $\Delta \geq 0$. Equality means that the flow is stationary from the first moment: the body was already Hanner. The classification of equality is not deferred to a second paper. It is a description of the attractors.

8. The node

Sections 3, 4, and 7 are closed. Sections 5 and 6 are closed only under the following statement.

Node. For every simple centrally symmetric polytope K in John position, there exists a probability measure μ_K on the sphere that simultaneously reproduces the surface-area measure of the body and the surface-area measure of the polar, in the sense of Section 5.

Without this measure, the identity $\delta \log P = \int (\eta + \eta^\circ) d\mu$ is not an identity, the filter is undefined, and monotonicity of Δ is not proved. Until the measure is constructed, the Mahler conjecture is not proved by this approach. The note records the node as the single open analytic point, not as a rhetorical caveat.

An external check of the contour, already separated in family 087 from the text of the manuscript, carries over unchanged. If the contour is closed, the symplectic width of the interior of $K \times K^\circ$ equals 4. A discrepancy from 4 means that the measure reproduced only one marginal. The filter is obliged to reject such a step.

9. The general case

Dropping $K = -K$ does not require a separate conical estimate. The position of the polar relative to an interior point becomes a variable of the flow. The only attractor with zero defect is the simplex. The constant changes to $(n+1)^{n+1}/(n!)^2$, because the homogeneous lift of the body is an orthant cone of dimension $n+1$, not a cube. The relation $\chi_C(v) \chi_{\{C^*\}}(u) \geq 1$ is the value of the same defect on the lift, not a new observable. This removes the fork in the trace at which the direct cone and the dual section lived on incompatible scales.

10. How the approach differs

The slab contour of family 087 closes the phase by a boundary law with triangular real part, a flux of the complex Hessian, and a packing of admissible simplices in the polar. Two sums compress into the bound. The approach of this note does not use that law. Closure is moved from the complex structure of the slabs to a moment measure on face–normal pairs. The bound is obtained not by comparing sums, but by the fact that the defect cannot grow on the way to the unique zero attractor.

For the search itself the difference is operational. A branch in which $|K|$ and $|K^\circ|$ are estimated separately is cut off before the text is unfolded: it has no single observable for the product. A branch is accepted only if it exhibits a measure with two marginals and non-increase of Δ . The main expenditure of the trajectory then goes not to holding defective sectors, but to constructing one node. That is the shift Invariant Becoming Theory introduces into the architecture of a proof: the state is a pair of form and defect, a step is allowed only if the defect does not grow, and recoherence is required at the junction of the body and the polar.

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Status. The note proposes an approach and carries it to a single analytic node. It does not claim that the symmetric Mahler conjecture has been proved.